

Daily Logic Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. Negate: “ n is even.”

Answer: “ n is odd.”

Method: For an integer n , “not even” is exactly “odd” — the two properties partition the integers.

Investigate further: Negation of a single property on integers is easy when the property has exactly one alternative; Section B negates compound statements where it’s less automatic.

2. Negate: “ $x > 5$.”

Answer: “ $x \leq 5$.”

Method: The negation of a strict inequality $x > 5$ is the non-strict opposite direction, $x \leq 5$.

Investigate further: Careless negation often drops the “or equal to” — $x < 5$ is a common wrong answer, since it forgets $x = 5$.

3. Negate: “ n is divisible by 3 and by 5.”

Answer: “ n is not divisible by 3, or n is not divisible by 5.”

Method: De Morgan’s law: $\text{not}(P \text{ and } Q) = (\text{not } P) \text{ or } (\text{not } Q)$.

Investigate further: The connective flips from “and” to “or” — this is the single most common negation mistake (see B9).

4. Negate: “ $x = 2$ or $x = 3$.”

Answer: “ $x \neq 2$ and $x \neq 3$.”

Method: De Morgan’s law: $\text{not}(P \text{ or } Q) = (\text{not } P) \text{ and } (\text{not } Q)$.

Investigate further: The mirror image of A3: “or” flips to “and.”

5. Negate: “For all integers n , $n^2 \geq 0$.”

Answer: “There exists an integer n such that $n^2 < 0$.”

Method: Negating $\forall n, P(n)$ gives $\exists n, \text{not } P(n)$; here $P(n)$ is “ $n^2 \geq 0$ ” and $\text{not } P(n)$ is “ $n^2 < 0$ ”.

Investigate further: The negation here is a statement that happens to be false (no such n exists) — negation is a purely mechanical operation independent of whether the result is true.

6. Negate: “There exists an integer n such that $n^2 = 2$.”

Answer: “For all integers n , $n^2 \neq 2$.”

Method: Negating $\exists n, P(n)$ gives $\forall n, \text{not } P(n)$; here $P(n)$ is “ $n^2 = 2$ ” and not $P(n)$ is “ $n^2 \neq 2$ ”.

Investigate further: Unlike A5, this negation is true: no integer squares to 2, since $1^2 = 1 < 2 < 4 = 2^2$.

7. True or false: “The negation of a true statement is always false.”

Answer: True.

Method: This is the fundamental law of non-contradiction: a statement and its negation can never both be true, nor both be false.

Investigate further: This is why proof by contradiction (Day 7’s toolkit) works at all: showing a statement’s negation leads to nonsense forces the original to be true.

8. State the contrapositive of: “If n is even, then n^2 is even.”

Answer: “If n^2 is odd, then n is odd.”

Method: The contrapositive of “if P then Q ” is “if not Q then not P ”: negate both clauses and swap their order.

Investigate further: Compare with Day 1’s converse of the same statement (“if n^2 even then n even”) — contrapositive and converse are different operations that can coincidentally share a truth value here, but not in general (see A10).

9. True or false: “The contrapositive of a statement is always logically equivalent to the original.”

Answer: True.

Method: “If P then Q ” is false only when P is true and Q is false; “if not Q then not P ” is false only when not Q is true and not P is false, i.e. Q is false and P is true — the exact same case. They agree in every case.

Investigate further: This is why proof by contrapositive (this sheet’s headline tool) is a fully legitimate proof method, not just a rephrasing trick.

10. True or false: “The contrapositive of a statement is the same as its converse.”

Answer: False.

Method: The converse of “if P then Q ” is “if Q then P ” (swap only); the contrapositive is “if not Q then not P ” (negate and swap). A2’s negation example already shows negating changes the statement, so these are generally different operations.

Investigate further: The contrapositive is always equivalent to the original (A9); the converse generally is not (Day 1’s whole toolkit) — confusing the two is the single most common logic error on TMUA Paper 2.

Section B Manipulation Drills

1. Negate: “If n is prime, then n is odd.”

Answer: “ n is prime and n is not odd (i.e. n is prime and n is even).”

Method: The negation of “if P then Q ” is “ P and not Q ”, never another “if...then” statement.

Investigate further: This negation happens to be satisfiable ($n = 2$), which is exactly why the original statement is false as stated (it should read “if n is prime and $n > 2$, then n is odd”).

2. Negate: “For all $x > 0$, $x^2 > 0$.”

Answer: “There exists $x > 0$ such that $x^2 \leq 0$.”

Method: Negating $\forall x > 0, P(x)$ gives $\exists x > 0$, not $P(x)$; here not $P(x)$ is “ $x^2 \leq 0$ ”. Note the domain restriction “ $x > 0$ ” is not itself negated, since it is part of the quantifier’s scope, not the statement being quantified.

Investigate further: This negation is unsatisfiable (no positive x has $x^2 \leq 0$), matching that the original was true.

3. Negate: “ n is a multiple of 4 or n is a multiple of 6.”

Answer: “ n is not a multiple of 4, and n is not a multiple of 6.”

Method: De Morgan’s law: $\text{not}(P \text{ or } Q) = (\text{not } P) \text{ and } (\text{not } Q)$.

Investigate further: $n = 5$ satisfies this negation (a multiple of neither), confirming the original “or” statement is not universally true.

4. State the contrapositive of: “If $x > 2$, then $x^2 > 4$.”

Answer: “If $x^2 \leq 4$, then $x \leq 2$.”

Method: Negate both clauses of “if $x > 2$ then $x^2 > 4$ ” and swap: hypothesis becomes “not $x^2 > 4$ ”, i.e. $x^2 \leq 4$; conclusion becomes “not $x > 2$ ”, i.e. $x \leq 2$.

Investigate further: The original “if $x > 2$ then $x^2 > 4$ ” is true for every real x (true whenever $x > 2$, vacuously true otherwise), so by A9 its contrapositive above must also be true for every real x — and it is: $x^2 \leq 4$ means $-2 \leq x \leq 2$, which certainly forces $x \leq 2$.

5. Prove “If n^2 is odd, then n is odd” by proving its contrapositive.

Answer: Proved via the contrapositive “if n is even, then n^2 is even” (Day 1, B2).

Method: The contrapositive of “if n^2 odd then n odd” is “if n even then n^2 even”, which was proved directly in Day 1 (B2): $n = 2k \Rightarrow n^2 = 4k^2 = 2(2k^2)$, even. Since the contrapositive is true and is logically equivalent to the original (A9), the original is also true.

Investigate further: This is the first time the sheet reuses an old proof unchanged, just relabelled as a contrapositive — exactly the payoff CONTRIBUTING.md’s toolkit design is built around.

6. Negate: “ n is prime and $n > 2$.”

Answer: “ n is not prime, or $n \leq 2$.”

Method: De Morgan’s law: $\text{not}(P \text{ and } Q) = (\text{not } P) \text{ or } (\text{not } Q)$.

Investigate further: Fixing B1's flawed statement: "if n is prime and $n > 2$, then n is odd" is actually true, and this negation is exactly what a contrapositive-by-contradiction argument would need to rule out.

7. Negate: "If it rains, then the ground is wet."

Answer: "It rains and the ground is not wet."

Method: The negation of "if P then Q " is " P and not Q " — never "if not P then not Q " (that's the inverse, a different and unrelated statement) and never "if Q then P " (the converse).

Investigate further: This everyday example is exactly why "if it rains, the ground is wet" being true does not mean "if it doesn't rain, the ground isn't wet" (a sprinkler could still wet it) — the inverse is not implied.

8. State the contrapositive of: "If n is a multiple of 9, then n is a multiple of 3."

Answer: "If n is not a multiple of 3, then n is not a multiple of 9."

Method: Negate both clauses and swap: hypothesis becomes " n not a multiple of 3", conclusion becomes " n not a multiple of 9".

Investigate further: This is Day 2's D5 statement S , now given its contrapositive directly — a useful cross-check against that answer's negation.

9. True or false: " $\text{not}(P \text{ and } Q)$ " is the same as " $(\text{not } P) \text{ and } (\text{not } Q)$ ".

Answer: False.

Method: The correct De Morgan law is $\text{not}(P \text{ and } Q) = (\text{not } P) \text{ or } (\text{not } Q)$, with "or", not "and". Concretely, with P : "2 is even" (true) and Q : "3 is even" (false): $\text{not}(P \text{ and } Q)$ is true (since P and Q is false), but $(\text{not } P) \text{ and } (\text{not } Q)$ is (false and true) = false — they disagree.

Investigate further: This is the single most common negation error — A3 and B3 already show the correct "and" $\rightarrow$ "or" flip in action.

10. True or false: " $\text{not}(P \text{ or } Q)$ " is the same as " $(\text{not } P) \text{ and } (\text{not } Q)$ ".

Answer: True.

Method: This is exactly De Morgan's law for "or": $\text{not}(P \text{ or } Q) = (\text{not } P) \text{ and } (\text{not } Q)$, matching A4's worked example.

Investigate further: Together, B9 and B10 are the two De Morgan laws — "and" negates to "or" and vice versa, always.

Section C Substitution & Structure

1. Negate: "For every prime p , p is odd or $p = 2$."

Answer: "There exists a prime p such that p is not odd and $p \neq 2$."

Method: Negate the quantifier ($\forall \rightarrow \exists$, negate the inner statement) then apply De Morgan to the "or": $\text{not}(p \text{ odd or } p=2) = (p \text{ not odd}) \text{ and } (p \neq 2)$.

Investigate further: This negation is unsatisfiable (every prime actually is odd or equals 2), matching that the original quantified statement is true.

2. Prove “If n^2 is even, then n is even” via its contrapositive.

Answer: Proved via the contrapositive “if n is odd, then n^2 is odd” (Day 1, B5).

Method: The contrapositive of “if n^2 even then n even” is “if n odd then n^2 odd”, proved directly in Day 1 (B5): $n = 2k + 1 \Rightarrow n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, odd. Since the contrapositive holds and is equivalent to the original, the original holds.

Investigate further: Combined with B5, both directions of “ n even $\iff n^2$ even” are now proved entirely by contrapositive, reusing Day 1’s two direct proofs without any new algebra.

3. Which is the correct negation of “If x is rational, then x^2 is rational”?

Answer: B) x is rational and x^2 is irrational.

Method: The negation of “if P then Q ” is always “ P and not Q ”, never another conditional. Here P : “ x rational”, Q : “ x^2 rational”, so the negation is “ x rational and x^2 irrational”, option B.

Investigate further: This negation is actually satisfiable: it would require x rational with x^2 irrational, which never happens (rationals are closed under squaring, Day 2’s D2) — so the original “if...then” is true, and this negation is correctly always false.

4. Negate: “There exists an integer n such that n is prime and n is even.”

Answer: “For all integers n , n is not prime or n is not even.”

Method: Negate the quantifier ($\exists \rightarrow \forall$, negate the inner statement) then apply De Morgan to the “and”: $\text{not}(n \text{ prime and } n \text{ even}) = (n \text{ not prime}) \text{ or } (n \text{ not even})$.

Investigate further: The original statement is true ($n = 2$), so this negation is false — it must fail specifically at $n = 2$, where both “not prime” and “not even” are false.

5. State and prove the contrapositive of: “If n is a multiple of 4, then n^2 is a multiple of 16.”

Answer: Contrapositive: “If n^2 is not a multiple of 16, then n is not a multiple of 4.”
Proved (via the original): writing $n = 4k$ gives $n^2 = 16k^2$, a multiple of 16.

Method: Prove the original directly: if $n = 4k$, then $n^2 = (4k)^2 = 16k^2$, manifestly a multiple of 16. Since the original is true and the contrapositive is logically equivalent to it (A9), the contrapositive is automatically also true — no separate proof of the contrapositive’s contrapositive form is needed.

Investigate further: Proving the easier direction (here, the original, via direct substitution) and getting the harder-looking direction (the contrapositive) for free is the entire point of the equivalence in A9.

6. True or false, with justification: “If a statement’s contrapositive is false, the original statement must also be false.”

Answer: True.

Method: By A9, a statement and its contrapositive always share the same truth value. If the contrapositive is false, the original — being logically equivalent — must be false too.

Investigate further: This is the contrapositive analogue of A7: just as a true statement’s negation is always false, a false contrapositive always forces a false original, with no exceptions.

7. Negate: “There exists a positive integer n such that n is not a multiple of any prime less than 10.”

Answer: “For every positive integer n , n is a multiple of some prime less than 10.”

Method: Negate the quantifier ($\exists \rightarrow \forall$) and negate the inner statement: “not (not a multiple of any prime less than 10)” simplifies (double negation) to “a multiple of some prime less than 10”.

Investigate further: This negation is false: $n = 1$ is a positive integer that is not a multiple of any prime, so the original existential statement (true at $n = 1$) has a genuinely false negation, as it must.

8. Prove “If n is not a multiple of 3, then n^2 is not a multiple of 3” via its contrapositive.

Answer: Proved via the contrapositive “if n^2 is a multiple of 3, then n is a multiple of 3” (Day 1, C1’s converse direction).

Method: The contrapositive of “if $3 \nmid n$ then $3 \nmid n^2$ ” is “if $3 \mid n^2$ then $3 \mid n$ ”. Since 3 is prime, if $3 \mid n^2$ then 3 must divide one of the two factors in $n \times n$, i.e. $3 \mid n$. This is exactly Day 1’s C1 discussion of the converse of “ $3 \mid n \Rightarrow 9 \mid n^2$ ” being true because 3 is prime.

Investigate further: This “prime divides a square $\Rightarrow$ prime divides the number” fact is the seed of unique factorisation, reused constantly in Number Theory.

Section D Challenge

1. Given the statement S : “If n is a multiple of 4, then n is even.” Which of the following is the contrapositive of S ?

Answer: A) If n is not even, then n is not a multiple of 4.

Method: Negate both clauses of S and swap: hypothesis “not even” (from negating the conclusion), conclusion “not a multiple of 4” (from negating the hypothesis). Option B is the converse (swap only), option C is the inverse (negate only, no swap), option D is an unrelated distractor.

Investigate further: All three wrong options correspond to a named, distinct operation (converse, inverse, nonsense) — recognising which named trap each wrong answer represents is the actual TMUA skill being tested here.

2. Prove by contrapositive: “If $n^2 - 2n$ is odd, then n is odd.”

Answer: Proved via the contrapositive “if n is even, then $n^2 - 2n$ is even.”

Method: Contrapositive: if n is even, write $n = 2k$; then $n^2 - 2n = n(n - 2) = 2k(2k - 2) = 4k(k - 1)$, manifestly even. Since the contrapositive holds and is equivalent to the original (A9), the original holds.

Investigate further: Direct proof would require assuming $n^2 - 2n$ is odd and deducing n ’s parity from an odd product $n(n - 2)$ — messier than simply proving the even case forces an even product, which is exactly why contrapositive is the natural tool here.

3. On an island, every inhabitant either always tells the truth or always lies. You meet two people, A and B . A says: “At least one of us is a liar.” Determine, with justification, whether A and B are truth-tellers or liars.

Answer: A is a truth-teller, B is a liar.

Method: Suppose A is a liar. Then A ’s statement “at least one of us is a liar” is false, meaning neither

A nor B is a liar — contradicting the assumption that A is a liar. So A must be a truth-teller, and A 's statement is true: at least one of A, B is a liar. Since A is not a liar, B must be.

Investigate further: This is a proof by contradiction in miniature (Day 7's toolkit, previewed here): assuming A is a liar and deriving a direct contradiction to that assumption forces the opposite conclusion.

4. Determine, with proof via contrapositive, whether the following is true: “If n^3 is even, then n is even.”

Answer: True; proved via the contrapositive “if n is odd, then n^3 is odd.”

Method: Contrapositive: if n is odd, write $n = 2k + 1$; then $n^3 = (2k + 1)^3$ is a product of three odd numbers, hence odd (odd $\times$ odd $\times$ odd = odd). Since the contrapositive holds and is equivalent to the original (A9), the original is true.

Investigate further: The same contrapositive pattern extends to any odd power: “if n^m is even then n is even” for any positive integer m , since n odd forces every power of n to stay odd.

5. Statement S : “If n is a multiple of 15, then n is a multiple of 3 and a multiple of 5.” (a) State the contrapositive of S . (b) Is “ n is a multiple of 15” necessary and sufficient for “ n is a multiple of 3 and a multiple of 5”? (c) Negate S directly (without first simplifying via contrapositive).

Answer: (a) “If n is not a multiple of 3 or not a multiple of 5, then n is not a multiple of 15.” (b) Yes, necessary and sufficient. (c) “ n is a multiple of 15, and (n is not a multiple of 3 or n is not a multiple of 5).”

Method: (a) Negate both clauses of S and swap; negating “ n is a multiple of 3 and a multiple of 5” gives, by De Morgan, “not a multiple of 3 or not a multiple of 5”. (b) Sufficient: $15 \mid n \Rightarrow 3 \mid n$ and $5 \mid n$ directly. Necessary: since $\gcd(3, 5) = 1$, $\text{lcm}(3, 5) = 15$, so $3 \mid n$ and $5 \mid n$ together force $15 \mid n$. Both directions hold. (c) Negating S itself (“ P and not Q ” form): P is “ n multiple of 15”, not Q is the De Morgan negation from part (a)'s inner clause, giving “ n multiple of 15, and (n not multiple of 3 or n not multiple of 5)”.

Investigate further: Part (c)'s negation is unsatisfiable (any multiple of 15 is automatically a multiple of both 3 and 5), consistent with S being true — three days of toolkit (contrapositive, necessary/sufficient, negation) converging on the same underlying fact from three different angles.